B

1. (a) appliances sold

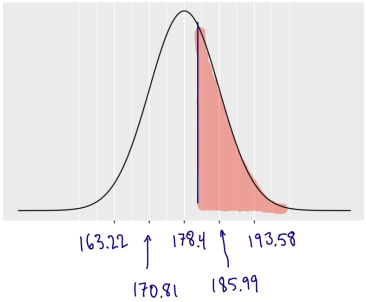
   (b) Whether the appliance is black or not

   (c) $H_0: p = 0.1$ vs. $H_a: p < 0.1$

2. (a) II (b) IV (c) I (d) III

4. (a) 1 - pnorm(181, 178.4, 7.59)

   (b)



   (c) qnorm(0.74, 178.4, 7.59)

   (d) $\sigma = 7.59$ and so $SE_{\bar{x}} = \frac{7.59}{\sqrt{16}} = 1.8975$

   pnorm(181, 178.4, 1.8975)

5. (a) μ is the average weight of these oranges considering the entire population.

   (b) qt(0.945, 24)

   (c) $123.4 \pm (1.65964)\frac{7.6}{\sqrt{25}} = 123.4 \pm (1.65964)(1.52) = 123.4 \pm 2.52265$

   or $(120.88, 125.92)$

   (d) Option (iii)

6. (a) $\hat{p} = \frac{43}{96} \doteq 0.44792$

   (b) It is valid, since $np = (96)(0.5) = 48$ and $n(1-p) = (96)(0.5) = 48$. Both are at least 5.

   (c) $Z = \frac{\hat{p} - p}{SE_{\hat{p}}} = \frac{0.4479 - 0.5}{\sqrt{(0.5)^2/96}} = \frac{-0.0521}{0.05103} \doteq -1.021$

   (d) pnorm(-1.021)

   (e) Since the p-value is larger than $\alpha = 0.05$, we fail to reject the null hypothesis. Our data is consistent with the possibility that half of city residents support having no restrictions.